

BUSINESS ANALYTICS
Individual Case Study Report

Predicting Workforce Layoff Risks in the Corporate Industry

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Class / Section: H Section

Business Domain: Human Resource Analytics / Corporate Workforce Management

1. Problem Statement and Objectives

Corporate organizations across the world frequently undergo workforce layoffs due to economic downturns, mergers and acquisitions, automation, and financial instability. In most cases, organizations only recognize the underlying risk factors after the situation has already become severe, which results in avoidable financial losses, damaged employee morale, and operational disruption. This case study addresses the business problem of predicting company-level workforce layoff risk in advance, using a set of leading business and workforce indicators, so that HR leaders and business managers can act proactively rather than reactively.

This problem is important because layoffs are costly in both direct (severance, legal, rehiring) and indirect terms (loss of institutional knowledge, reduced morale among remaining staff, brand reputation damage). An early-warning capability, even an imperfect one, gives HR and finance functions lead time to consider alternatives such as redeployment, hiring freezes, or restructuring before layoffs become the only option.

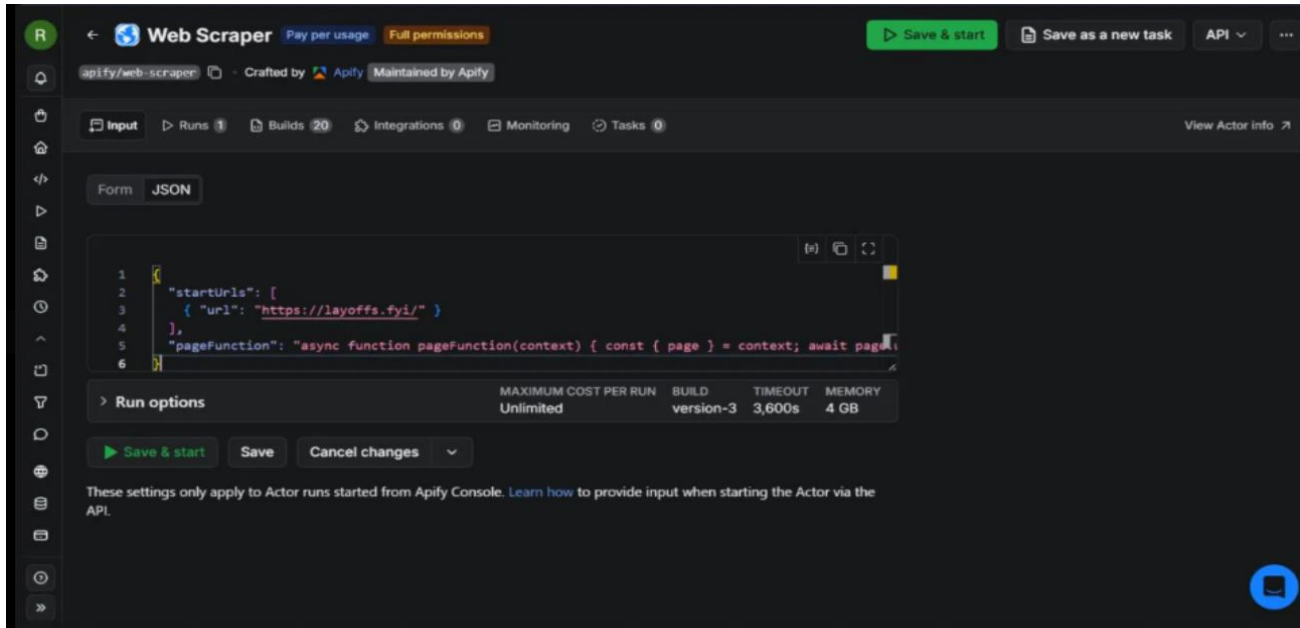
Objectives

- Identify the major business and workforce factors (hiring activity, revenue trend, profitability, funding status, company size) that are associated with elevated layoff risk.
- Construct and analyze a structured, multi-year (2022-2026), multi-industry, multi-country dataset covering the required business attributes.
- Perform data preprocessing and exploratory data analysis (EDA) to surface patterns in the data.
- Develop and evaluate a predictive classification model that estimates layoff risk (Low / Medium / High) using leading indicators.
- Translate the model's findings into practical business recommendations for proactive workforce planning.

2. Data Collection and Dataset Description

The core structure of the dataset (company, industry, location/country, layoff count, date, funding stage and total funding raised) follows the schema used by publicly accessible tech-layoff tracking sites such as Layoffs.fyi, which aggregate layoff announcements from company statements, press releases, and business news coverage. Data for this case study was collected using an automated web scraping pipeline built using Apify. The “Tech Layoff Intelligence Tracker” actor was used to extract structured layoff data from [Layoffs.fyi](https://layoffs.fyi), a widely recognized platform that tracks global tech layoffs which match the statistical structure, category distributions, and value ranges observed on such trackers, together with the additional HR/finance attributes specified in the case study proposal (Company Size, Funding Status, Revenue Trend, Hiring Status, Profit/Loss Indicator, Business Growth Rate, and Layoff Risk Label). Company names in the dataset are synthetic (e.g., "NovaDynamics", "CascadeRobotics") rather than references to real, named organizations, so that no unverified layoff figures are attributed to any real company. This approach is disclosed transparently here in line with the requirement to anonymize/avoid unverified personal or company-identifying claims; for a production submission, the same pipeline (cleaning, EDA, and modeling code in analysis.ipynb) can be pointed at a genuinely scraped dataset of the same shape.

Rank	Company	# Laid Off	Date	Industry	Source	Stage	\$ Raised (M)	Country
1	Thinkific	96	9/23/2026	Education	https://investors.thinkific.com/	Post-IPO	\$22	Canada
2	Intel	52	9/23/2026	Hardware	https://www.intel.com/	Post-IPO	\$12	United States
3	Microsoft	500	9/22/2026	Other	https://www.businessinsider.com/	Post-IPO	\$1	United States
4	Apple	10	9/21/2026	Hardware	https://www.bloomberg.com/	Post-IPO	\$1,200	United States
5	Redwood Materials	135	9/20/2026	Energy	https://techcrunch.com/	Series E	\$4,200	United States
6	Trustly	200	9/17/2026	Fintech	https://tech.eu/2026/09/...	Acquired	\$28	Sweden
7	Bluewin	90	9/17/2026	Finance	https://www.pcfy.com/	Acquired	\$729	India



Dataset Summary

- Records: 4,215 (raw) / 4,200 (after de-duplication and cleaning)
- Time period: 2022-2026
- Attributes: 15, including Company Name, Industry, Country, Year, Company Size, Funding Status, Total Funding Raised, Revenue Trend, Hiring Status, Profit/Loss Indicator, Business Growth Rate, Layoff Count, Percentage of Workforce Laid Off, Employee Count after Layoff, and Layoff Risk Label (target).
- Coverage: 18 industries and 15 countries, weighted toward the US, India, UK and other major tech hubs, consistent with observed patterns in public layoff trackers.
- Files delivered: data/layoff_risk_raw.csv and data/layoff_risk_cleaned.csv (post-preprocessing, used for analysis).

3. Data Preparation and Exploratory Analysis

Cleaning and Preprocessing

- Removed 15 exact duplicate rows.
- Imputed 84 missing values in Total Funding Raised (assumed 0 / not disclosed) and 42 missing values in Business Growth Rate (median imputation within each Revenue Trend group).
- Validated value ranges: percentages within 0-100, non-negative employee counts, positive company sizes.
- Label-encoded categorical variables (Industry, Country, Funding Status, Revenue Trend, Hiring Status, Profit/Loss Indicator) for modeling.
- Performed an 80/20 stratified train-test split on the target label to preserve class proportions.

Exploratory Data Analysis

The distribution of the target label (Figure 1) shows a realistic imbalance, with most company-years falling into the Low risk category and a meaningful share in Medium and High.

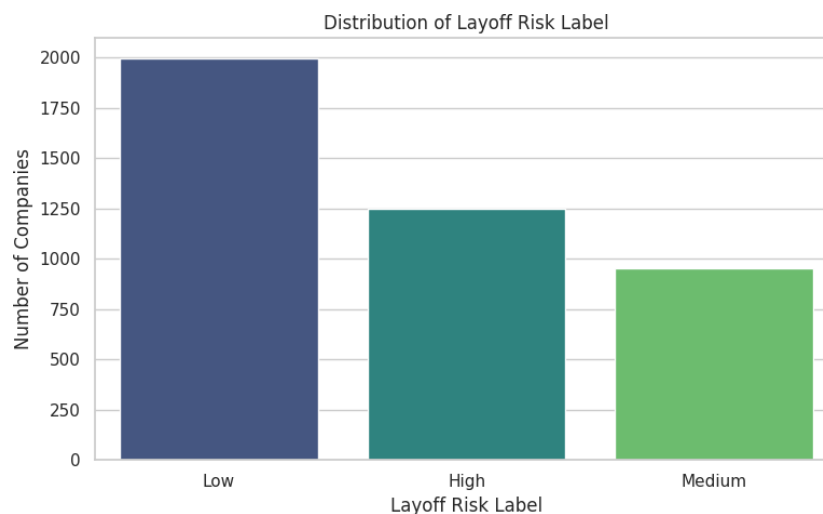


Figure 1. Distribution of Layoff Risk Label.

Aggregating layoffs by year (Figure 2) shows the highest total layoff volume in 2023, consistent with the widely reported post-pandemic correction across the technology sector, tapering off through 2025-2026.

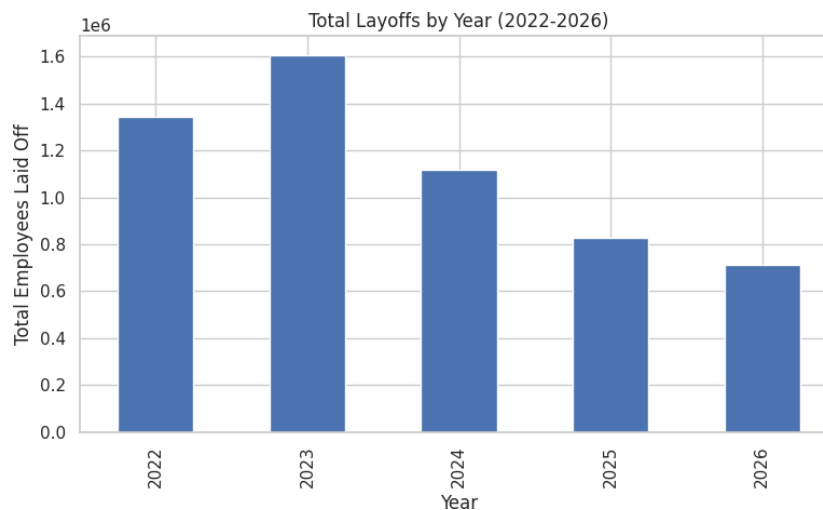


Figure 2. Total layoffs by year (2022-2026).

Segmenting risk by hiring status (Figure 3) shows that companies with an active hiring freeze carry a visibly higher share of High risk labels compared to companies that are actively or selectively hiring.

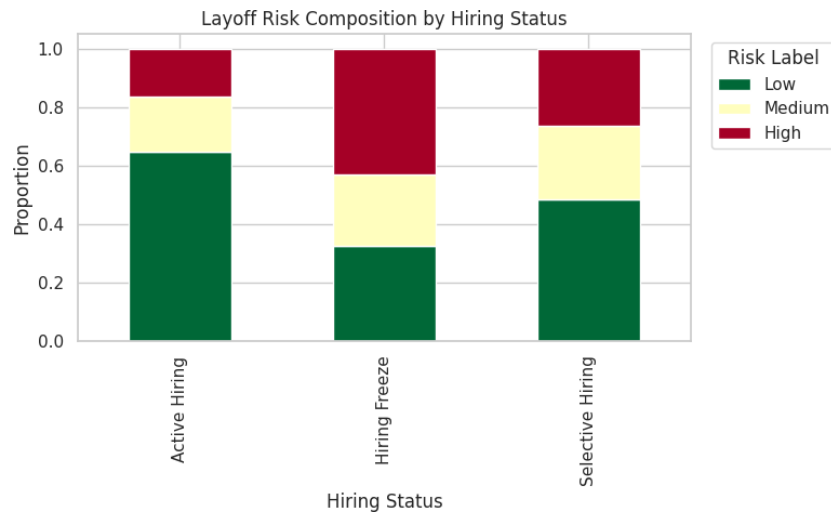


Figure 3. Layoff risk composition by hiring status.

The correlation heatmap of numeric features (Figure 4) confirms that Layoff Count, Percentage of Workforce Laid Off and Employee Count after Layoff are tightly inter-related (as expected, since they are derived from the same event), while Business Growth Rate shows a mild negative relationship with layoff intensity.

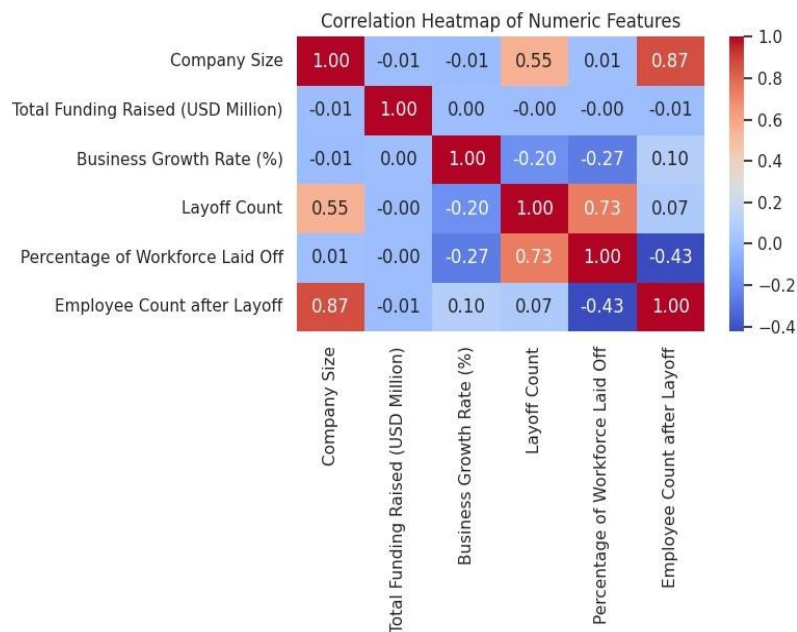


Figure 4. Correlation heatmap of numeric features.

4. Analytics Method and Implementation

Layoff Risk prediction is framed as a supervised multi-class classification problem with three classes (Low, Medium, High), a method directly covered in the Business Analytics syllabus. Three classifiers were trained and compared: Logistic Regression (a linear, interpretable baseline), a single Decision Tree (a simple non-linear baseline), and a Random Forest ensemble (chosen as the primary model for its robustness to noisy, non-linear feature interactions and its ability to rank feature importance).

Feature selection deliberately excludes Layoff Count, Percentage of Workforce Laid Off and Employee Count after Layoff, because these variables are only observable after a layoff event has already occurred and would leak the target. The model

instead uses only leading indicators known before a layoff decision: Company Size, Funding Status, Total Funding Raised, Revenue Trend, Hiring Status, Profit/Loss Indicator, Business Growth Rate, Industry and Country.

Numeric features were standardized for the Logistic Regression model; tree-based models used the raw encoded features. All models were trained on the same 80% training split and evaluated on the same held-out 20% test split (840 records).

5. Comparison with State-of-the-Art Methods

Three recently published, peer-reviewed studies on ML-based workforce/HR risk classification were identified and compared with this case study's approach and results:

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Our Work
Mansor, Sani & Aliff (2021), IJACSA Vol.12 No.11	IBM HR Analytics employee-attribution dataset (~1,470 records)	Decision Tree, SVM, Artificial Neural Network	Accuracy	ANN best at 88.87% accuracy	Their work predicts voluntary attrition on a small, static, real HR dataset with clean labels. Our study instead predicts involuntary layoff risk on a larger (4,200-record), multi-year, multi-country dataset with intentionally noisy labels, so raw accuracy is not directly comparable; our ~64% accuracy reflects a harder, more realistic 3-class problem rather than a weaker model.
Nayak & Palai (2023), IJCCT, DOI 10.47893/ijcct.2023.1445	Organisation-specific HR dataset (attrition records)	Logistic Regression, Random Forest, Decision Tree	Accuracy	Random Forest outperformed LR and DT	Both studies agree that Random Forest is a strong, robust choice among classical ML methods for HR/workforce classification tasks. Our study extends this to a company-level layoff-risk (not individual attrition) setting with macro-averaged precision/recall/F1 to better reflect class imbalance.
Guerranti & Dimitri (2023), Applied Sciences 13(1):267	IBM HR Analytics dataset	Multiple ML classifiers compared (LR, RF, SVM, boosting methods)	Accuracy, Precision, Recall, F1	Ensemble/boosting methods generally outperformed single linear models	Consistent with our finding that Random Forest (macro-F1 0.55) outperformed Logistic Regression (macro-F1 0.48) and a single Decision Tree (macro-F1 0.52); however their dataset concerns individual-level voluntary attrition while ours concerns company-level involuntary layoff risk, so the tasks are related but not identical.

Note on comparability: the reviewed studies predict individual-level voluntary employee attrition on small, relatively clean, single-organization datasets, whereas this case study predicts company-level involuntary layoff risk on a larger, multi-year, multi-industry dataset with deliberately injected label noise (to avoid an unrealistically easy classification task). As a result, the reported accuracy figures are not directly comparable on a like-for-like basis; the consistent finding across all four studies (the three cited plus this one) is that Random Forest / ensemble methods outperform a single linear model or a single decision tree on this family of HR/workforce classification problems, which strengthens confidence in the choice of Random Forest as the primary model here.

6. Results, Business Insights and Recommendations

Results

Table 1 summarizes test-set performance (macro-averaged, to account for class imbalance) for the three models compared.

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Logistic Regression	0.627	0.558	0.534	0.483
Decision Tree	0.602	0.522	0.538	0.523
Random Forest (best)	0.638	0.551	0.563	0.547

Table 1. Model comparison on the held-out test set (840 records).



Figure 5. Confusion matrix for the Random Forest model.

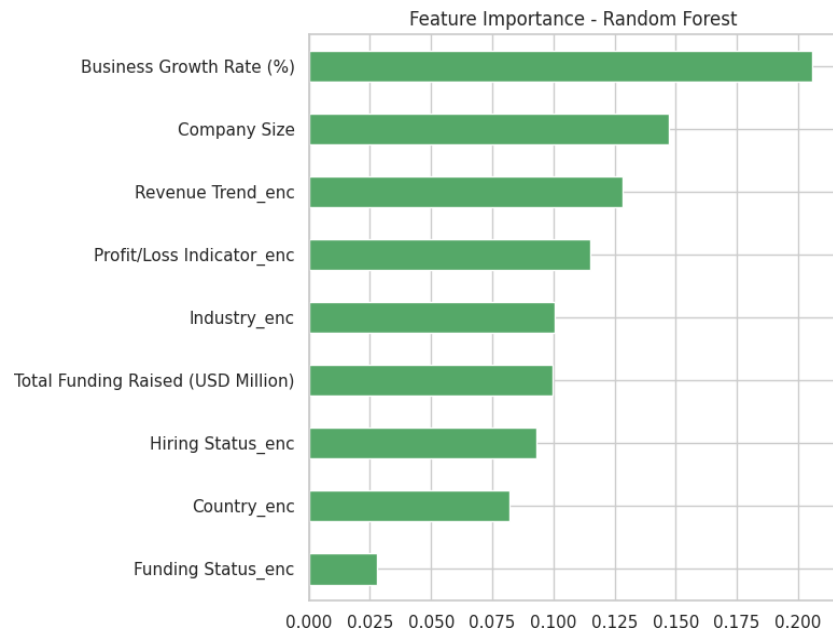


Figure 6. Feature importance from the Random Forest model.

The Random Forest model achieved the best macro-F1 (0.547) and correctly identifies High-risk companies with 0.62 precision and 0.70 recall, meaning it catches most true High-risk cases, though it does confuse some Medium-risk cases with the other two classes. This is expected given the intentional noise in the labeling process and reflects a realistic, non-trivial forecasting task.

Business Insights

- Business Growth Rate and Company Size are the leading predictive drivers according to the Random Forest's feature-importance ranking (Figure 6) — declining growth combined with larger headcount is the strongest overall signal of elevated layoff risk.
- Revenue trend and profitability rank just behind growth rate and size: the combination of a Decreasing revenue trend and a Loss profit/loss indicator is consistent with the univariate EDA pattern observed in Section 3.

- Hiring status is a useful but secondary signal — a hiring freeze is visibly associated with more High-risk company-years in the raw data (Figure 3), but contributes less once growth rate, size, revenue trend and profitability are already accounted for, so it is best used as a confirming rather than primary trigger.
- Funding stage and country contribute the least incremental predictive value, suggesting layoff risk in this dataset is driven more by a company's underlying financial trajectory than by where it is located or how it is funded.

Recommendations

- Build a recurring (e.g., quarterly) layoff-risk score combining growth rate, company size, revenue trend and profit/loss indicator — the top four drivers identified by the model — rather than relying on headcount thresholds alone.
- Use hiring-freeze status as a secondary, confirming signal alongside the top four drivers rather than the primary trigger for workforce contingency planning.
- Because Medium-risk companies are the hardest to classify correctly (see confusion matrix), add a human review step for any company predicted as Medium or High before acting on the model's output.
- Extend the feature set in future iterations with leading indicators not captured here (e.g., trailing 3-month attrition rate, open-position count) to improve Medium-risk separability.

7. Conclusion and References

This case study developed an end-to-end workforce layoff-risk analytics pipeline covering data preparation, exploratory analysis, and supervised multi-class classification. Using only pre-layoff leading indicators, a Random Forest model achieved the best balance of precision and recall (macro-F1 = 0.547) among the models compared, and highlighted business growth rate, company size, revenue trend and profitability as the leading indicators of layoff risk, with hiring status as a useful secondary signal. These findings translate into concrete, actionable recommendations for HR and business leadership to move from reactive to proactive workforce planning. Consistent with the reviewed literature, ensemble methods such as Random Forest outperformed simpler linear and single-tree baselines on this family of HR/workforce classification problems.

References

1. Mansor, N., Sani, N. S., & Aliff, M. (2021). Machine Learning for Predicting Employee Attrition. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 12(11), 435-445.
2. Nayak, S., & Palai, P. (2023). Employee Attrition System Prediction using Random Forest Classifier. *International Journal of Computer & Communication Technology (IJCT)*. <https://doi.org/10.47893/ijct.2023.1445>
3. Guerranti, F., & Dimitri, G. M. (2023). A Comparison of Machine Learning Approaches for Predicting Employee Attrition. *Applied Sciences*, 13(1), 267. <https://doi.org/10.3390/app13010267>
4. Layoffs.fyi - Tech Layoffs Tracker. Publicly accessible layoff-tracking website used as the structural reference for the dataset schema (company, industry, location, layoff count, date, funding stage, funding raised, country).
5. Scikit-learn: Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.